

A Summer Industry Internship-I
Report on
JAVA FULL STACK DEVELOPMENT

During III Year I Semester

A.Y: 2025-26

Submitted to

The Department of Computer Science and Engineering

In partial fulfillment of the requirement for

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BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING

Submitted by

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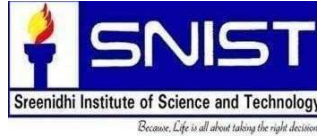
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CERTIFICATE

This is to certify that **this Summer Industry Internship – I** Report on Internship carried out in “**CODSOFT**”, submitted by **Himanshu Bijja (23311A05P5)** ,in the year 2025 in partial fulfillment of the academic requirements of Jawaharlal Nehru Technological University for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is a bonafide work in summer industry internship that has been carried out **during III D Tech CSE I semester**, will be evaluated in **III D Tech CSE I Semester**. This report has not been submitted to any other institute or university for the award of any degree.

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CERTIFICATE

OF COMPLETION
PROUDLY PRESENTED TO

Himanshu Bijja

has successfully completed 4 weeks of a virtual internship program in

Web Development

with wonderful remarks at **CODSOFT** from 01/11/2025 to 30/11/2025.

We were truly amazed by his/her showcased skills and invaluable contributions to the tasks and projects throughout the internship.



A handwritten signature in black ink, appearing to read "Ravi An", positioned above a horizontal line.

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DECLARATION

I, **Himanshu Bijja (23311A05P5)** , student of **SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY, YAMNAMPET, GHATKESAR**, studying **IIIrd year Ist semester**, **COMPUTER SCIENCE AND ENGINEERING** solemnly declare that the **Summer Industry Internship-I Report**, is submitted to **SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY** for partial fulfillment for the award of degree of Bachelor of technology in **COMPUTER SCIENCE AND ENGINEERING**.

It is declared to the best of our knowledge that the work reported does not form part of any dissertation submitted to any other University or Institute for award of any degree

Himanshu Bijja
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ACKNOWLEDGEMENT

I would like to express my gratitude to all the people behind the screen who helped me to transform an idea into a real application. I would like to express my heart-felt gratitude to my parents without whom i would not have been privileged to achieve and fulfill my dreams. We are grateful to our principal, **Dr. T. Ch. Siva Reddy**, who most ably runs the institution and has had the major hand in enabling me to do my project. I profoundly thank **Dr. K. Shirisha, Head of the Department of Computer Science & Engineering** who has been an excellent guide and also a great source of inspiration to our work. We would like to thank our Coordinator **Ms Pavani Bajjuri** for Summer Industry Internship-II for their technical guidance, constant guidelines, encouragement and support in carrying out my **Summer Industry Internship-I** on time at college. The satisfaction and euphoria that accompany the successful completion of the task would be great but incomplete without the mention of the people who made it possible with their constant guidance and encouragement crowns all the efforts with success. In this context, we would like to thank all the other staff members, both teaching and nonteaching, who have extended their timely help and eased my task.

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Abstract

The CodSoft Web Development Internship provides a hands-on learning experience designed to build strong fundamentals in modern web technologies. Interns work with HTML, CSS, JavaScript, and frameworks to develop responsive and visually appealing websites. The program includes real-time tasks such as creating landing pages, portfolio websites, and functional web applications that reflect industry standards. Through these projects, participants learn page structuring, UI styling, interactive components, and client-side logic implementation. The internship also focuses on problem-solving, clean coding practices, and understanding user-centric design. By the end, interns gain practical development experience and are well-prepared for real-world web development roles. Interns also gain exposure to version control using Git and learn how to deploy websites on popular hosting platforms. The internship encourages creativity and innovation by allowing participants to experiment with modern tools and libraries. By collaborating on tasks and building real-world projects, interns strengthen their technical confidence and professional readiness for the evolving web development industry.

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1 .INTRODUCTION

1.1About the Internship

My Summer Internship in **Java Full Stack Web Development** was an 4 week virtual training program designed to build strong foundational and practical skills required in full-stack application development. The internship combined **frontend, backend, database, and development workflow concepts**, giving me an end-to-end understanding of how modern web applications are built, tested, and deployed.

The internship bridged the gap between theoretical knowledge and real-world implementation. It provided me with a strong technical foundation and practical exposure, enabling me to understand how actual industry-level applications are designed, developed, optimized, and deployed. Throughout the program, I worked with essential technologies such as **HTML, CSS, JavaScript, Tailwind CSS, Git, GitHub, and Web Hosting platforms**, which collectively form the building blocks of a full-stack developer's skill set.

This internship also familiarized me with collaborative development workflows, communication patterns, problem-solving techniques, and real-time debugging practices used in professional environments. Daily interactions with mentors, structured tasks, and project-oriented modules helped me gain clarity on how software teams operate and how responsibilities are distributed across development cycles.

Working with Frontend Technologies: HTML, CSS, JavaScript

A major portion of the internship was focused on developing strong frontend skills. I gained practical knowledge of creating visually appealing, structurally sound, interactive, and responsive web pages.

HTML – Structuring the Web Interface

HTML served as the backbone of all webpages I developed. During the internship, I learned:

- How to create clean, semantic HTML structures
- How to design sections, headers, footers, navigation bars, and forms
- How to ensure accessibility using proper tags
- The importance of structured layouts for SEO and readability
- Embedding multimedia (images, audio, video)
- Creating hyperlinks and connecting multiple pages

- Using form elements for user inputs

CSS – Designing and Styling Web Interfaces

CSS allowed me to transform plain HTML pages into visually appealing, user-friendly, and attractive designs. Topics I mastered included:

- Selectors, properties, and values
- Box model (padding, margin, border)
- Flexbox and Grid for layout alignment
- Utility classes and reusable styling patterns
- Media queries for responsive web design
- Creating consistency in buttons, typography, and color schemes
- Animations, transitions, and hover effects
- Designing mobile-first layouts

With these skills, I was able to design interfaces that were both visually consistent and optimized for multiple screen sizes.

JavaScript – Adding Interactivity

JavaScript was one of the most important technologies I learned during the internship, as it brought interactivity and logic to the webpages. I worked on:

- DOM manipulation (updating content dynamically)
- Event handling (clicks, inputs, forms, mouse actions)
- Form validation (checking inputs, errors, restrictions)
- Conditional rendering and decision-making
- Creating interactive menus, toggles, modals
- Arrays, objects, loops, and functions
- ES6 concepts like let/const, arrow functions, templates

These tasks helped me understand the role of JavaScript in improving usability and enhancing user experience.

Experience with Modern Tools & Practices

Version Control with Git & GitHub

During the internship, I used Git and GitHub for source code management.

I learned to:

- Create and manage repositories

- Work with branches
- Commit and push code changes
- Resolve merge conflicts
- Review and collaborate on code

This experience taught me the importance of version control in real-world projects and teamwork.

Responsive and Mobile-First Development

I was involved in building pages that adjust smoothly across different screen sizes. I worked with:

- Breakpoints and media queries
- Fluid layouts
- Scalable images and components

This gave me a strong foundation in responsive web design, which is essential for today's multi-device users.

Quality Assurance and Testing

Testing and debugging are essential for producing error-free and user-friendly applications. As part of the internship:

Functional Testing

I tested:

- User inputs
- Page navigation
- Button clicks
- Form submissions
- Error handling

Debugging Using DevTools

Using Chrome DevTools, I learned to:

- Inspect HTML & CSS
- Debug JavaScript issues
- Monitor console errors
- Simulate responsive views
- Improve performance

This experience improved my attention to detail and helped me identify and resolve issues quickly.

Team Collaboration & Communication

The internship involved continuous collaboration with mentors and peers. This improved my:

- Communication skills
- Teamwork and adaptability
- Professional etiquette
- Time management
- Task planning and execution

Participating in discussions, reviewing tasks, and interacting with team members helped me understand real-world work culture and project workflows.

Professional Growth & Takeaways

By the end of the internship, I gained:

- A strong foundation in frontend development
- Ability to build interactive and responsive webpages
- Experience in using Git for collaborative projects
- Improved analytical and problem-solving skills
- Exposure to real-world development environments
- Confidence to handle software development tasks independently

This internship strengthened my interest in full-stack development and motivated me to pursue more advanced concepts like backend development (Java, Spring Boot) and database management.

1.2 Plan of Training program

It has 12 modules

Module 1: Frontend Development – HTML

- Introduction to HTML
- Structure of web pages
- Elements, attributes, and tags
- Forms, tables, lists
- Page layout and content arrangement
- Integrating multimedia into webpages

Module 2:CSS

- Introduction to CSS and its syntax
- Selectors, properties, and values
- Box Model: margin, padding, border
- Layout techniques: Flexbox, Grid
- Responsive design principles
- Styling webpages for better user interface

Module 3: JavaScript

- Basics of JavaScript
- Variables, functions, and objects
- DOM manipulation
- Event handling
- Adding interactivity to webpages
- Form validation and basic dynamic features

Module 4: Tailwind CSS

- Introduction to Tailwind CSS
- Utility-first CSS approach
- Applying spacing, colors, and typography
- Flexbox and Grid using Tailwind classes
- Creating clean and responsive layouts

Module 5: Version Control – Git & GitHub

- Introduction to Git
- Git commands: init, add, commit, push
- Branching and merging
- Collaboration in GitHub

Module 6: Web Hosting

- What is web hosting
- Types of hosting
- Steps for deploying a website
- Hosting using cloud platforms

Module 7: Core Java

- OOP principles
- Java syntax ,Exception handling
- Collection framework ,File handling
- Multithreading ,Writing java programs

Module 8: Advanced Java

- JDBC
- Servlets and JSP
- JSTL
- MVC architecture

Module 9: Spring Framework

- Spring Boot fundamentals
- Spring MVC
- Spring Data JPA & Hibernate

Module 10: Database Module

- SQL basics
- MySQL, CRUD operations
- Joins, indexing, and views

Module 11: API testing

- API testing using Postman
- REST endpoint testing
- Response verification

Module 12: DevOps Basics

- Maven/Gradle
- Docker fundamentals ,CI/CD introduction
- Cloud deployment

1.3 Technology Learnt

1. HTML(Hyper Text Markup language)

HTML is the standard markup language used to create and structure content on the web.

Purpose:

- Structures web content using elements like headings, paragraphs, lists, links, and multimedia.
- Defines the building blocks of web pages.

Basic Structure:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Page Title</title>
  </head><body>
    <h1>Welcome</h1>
    <p>This is a paragraph.</p>

  </body>
</html>
```

Elements:

- Consist of opening and closing tags.
- Example: <p> and </p> for a paragraph.
- Some tags are self-closing, e.g., .

Attributes:

- Provide additional information about elements.
- Specified in the opening tag.
- Example: Visit Example.

Semantic Tags:

- Introduced in HTML5 for better structure and readability.
- Examples: <header>, <footer>, <article>, <section>.

Forms and Input:

- Allows user interaction through input fields, buttons, checkboxes, etc.

```
<form>
  <input type="text" name="username" placeholder="Enter name">
  <button type="submit">Submit</button>
</form>
```

Multimedia Integration:

- Embeds images, videos, and audio.
- Examples:

```

<video controls>
  <source src="video.mp4" type="video/mp4">
</video>
```

• Hyperlinks:

- Create navigable links between pages or within the same page. • Example: Click Here.

• Versions:

- HTML has evolved over time (HTML4, XHTML, HTML5).
- HTML5 is the current version with enhanced features like multimedia support and new semantic tags.

• Integration with Other Technologies:

- Works with CSS for styling.
- Works with JavaScript for dynamic behavior.

2.CSS (Cascading style Sheets)

CSS: A stylesheet language used to style and layout web pages, controlling the visual presentation of HTML elements.

Purpose:

- Separates content (HTML) from design (CSS).
- Enhances website appearance and user experience. Basic Syntax: property: value;

Selectors:

- Used to target HTML elements.
- Types:
 - Universal: * { }
 - Type: h1 { }
- Class.className{ }
- ID:#idname { }
- Attribute[type="text"]{ }

Types of CSS:

- Inline: Styles directly in HTML elements.
- Internal: Styles in a <style> tag in the HTML <head>.
- External: Styles in a separate .css file.

Box Model:

- Describes the structure of an element, including:
 - Content
 - Padding
 - Border
 - Margin

Positioning:

- Controls element placement.
- Values: static, relative, absolute, fixed, sticky.

Responsive Design:

- Ensures websites look good on all devices.
- Techniques include:

CSS Units:

- Absolute: px, cm, mm.
- Relative: %, em, rem, vh, vw.

CSS Frameworks:

- Prebuilt libraries to simplify styling.
- Examples: Bootstrap, Tailwind CSS, Bulma.

Animations and Transitions:

- Add dynamic effects to elements.

```
div {  
  transition: background-color  
  0.5s; } div:hover {  
  background-color: yellow; }
```

Flexbox:

- Provides a layout model for arranging items in a container.

```
display: flex; justify-  
content: center; Grid:
```

- A powerful system for creating complex layouts.

Custom Properties (CSS Variables):

- Define reusable values.

```
:root {  
  --main-color: #ff5733;  
} h1 { color: var(--  
maincolor);  
}
```

3.JavaScript

JavaScript: A lightweight, interpreted programming language used to create interactive and dynamic websites.

Purpose: Adds behavior to websites, enabling features like form validation, animations, and dynamic content updates.

Core Features:

- Supports Object-Oriented, Functional, and Procedural programming.
- Executes directly in the browser without compilation.

Basic Syntax:

Example: `console.log("Hello, World!");` Data Types:

Primitive: String, Number, Boolean, Null, Undefined, Symbol, BigInt.

Non-primitive: Object, Array, Function.

Variables: Declared using `var`, `let`, or `const`. Example:

```
let name = "John";
```

```
let name = "John"; const pi  
= 3.14;
```

Functions:

- Define reusable blocks of code

```
function    greet(name) {  
  return `Hello, ${name}`;  
}
```

Error Handling:

- Manages errors using try-catch blocks.

```
try    {  
  riskyOperation();    }  
catch (error) {  
  console.error(error);  
}
```

Frameworks and Libraries:

Popular libraries: jQuery, Lodash.

- Popular frameworks: React, Angular, Vue.
- ES6 and Modern JavaScript:

Event Loop:

- Manages execution of JavaScript code, especially asynchronous operations.
- Handles the call stack and callback queue.

4.TailWind CSS

- Tailwind CSS is a utility-first CSS framework for building custom designs quickly by using pre-defined utility classes directly in HTML.

Purpose:

- Simplifies the styling process by eliminating the need to write custom CSS.
- Encourages a consistent and reusable design system.

Key Features:

- Utility-First Classes: Provides low-level utility classes for margin, padding, colors, typography, etc.
- Customizability: Fully customizable via a configuration file.
- Responsive Design: Built-in support for responsive breakpoints.

```
<button class="bg-blue-500 hover:bg-blue-700 text-white font-bold py-2 px-4 rounded">
```

Click Me

```
</button>
```

Responsive Design:

- Tailwind uses breakpoint prefixes for responsive styles:

sm: Small screens (640px).

md: Medium screens (768px).

lg: Large screens (1024px).

5. Git & GitHub (Version Control System)

Git is used to track code changes ,and GitHub is used for storing and collaborating on projects.

Purpose:

- Manage versions of code
- Team collaboration
- Backup of projects

Basic Command:

git init

```
git add .  
git commit -m "message"  
git push
```

Elements:

Features:

- Branching.
- Merging
- Pull requests

Multimedia Integration:

- Used with ant project to track updates

6.Web Hosting

Hosting allows deploying websites and applications online..

Purpose:

- Make websites accessible on the internet
- Store files on a server

7.Core Java

Core Java covers fundamental programming concepts essential for backend development..

Purpose: Build logic based application and understand OOP concepts

Key Concepts:

- Classes and objects
- Constructors
- Exception Handling
- Collections
- Multithreading.

Example:

```
class Hello {  
    public static void main(String args[]){  
        System.out.println("Hello World");  
    }  
}
```

8.Advanced Java

Advanced Java focuses on building dynamic web applications.

Technologies:

- JDBC
- Servlets
- JSP
- JSTL
- MVC Architecture

Example:

```
Connection con = DriverManager.getConnection(url, user, pass);
```

9.Database

SQL is used to store, manage, and retrieve data.

Purpose:

- Manage application data
- Perform CRUD operations

Basic Queries:

```
CREATE TABLE students(id INT, name VARCHAR(50));  
INSERT INTO students VALUES(1,'John');  
SELECT * FROM students;
```

2.INTERNSHIP FEEDBACK

1. Collaboration and Teamwork:

The collaborative environment during the internship was one of the most valuable aspects of my experience.

As a team, we tackled various aspects of the project, including development, design, and testing.

Version Control: We used Git and GitHub for version control, which allowed us to work on the codebase simultaneously while tracking changes, resolving merge conflicts, and reviewing each other's work.

Task Management: Using tools like Jira for task management helped us stay organized and aligned. We broke the project into manageable tasks and tracked progress using sprints, ensuring that we met our goals on time.

Communication: Regular communication with the team was essential. I participated in daily standups where we discussed progress, challenges, and next steps. This improved my communication skills and allowed me to stay on the same page with the team.

2. Design and Tools Proficiency:

During the internship, I also had the opportunity to work on the design aspects of the project, which helped me gain proficiency in various design tools.

PowerPoint: I used PowerPoint to create presentations that summarized project updates, discussed challenges, and outlined future goals. This helped me learn how to communicate complex technical information in a simple and visually appealing manner.

Canva: I used Canva to create graphics for the platform. This included designing banners, promotional material, and simple UI components. The drag-and-drop interface made it easy for me to create professional-quality visuals, even as a beginner.

The ability to integrate design and technical work made me more versatile, equipping me with the skills needed to approach web development from both a technical and design perspective.

3. Testing and Quality Assurance:

An essential part of my internship experience was learning about the importance of quality assurance and black-box testing. Testing ensured that the platform functioned as expected and provided a smooth user experience.

Test Case Design: I was involved in creating test cases based on the functional requirements of the application. This included testing different user interactions such as form submissions, login flows, and content retrieval.

Functionality Testing: I ran black-box tests where I did not have access to the underlying code. Instead, I focused on testing the application's functionality, ensuring that all user-facing features worked as expected.

Performance Testing: I also contributed to performance testing by measuring how the platform performed under different conditions, like multiple users interacting with the site simultaneously.

Bug Reporting and Debugging: I used tools like Jira to report bugs and track fixes. This allowed me to work closely with the development team to ensure the platform was stable and functional.

4. Challenges Faced:

During my internship, I encountered several challenges that helped me grow both technically and professionally:

Balancing Frontend and Backend Work: The dual focus of full-stack development often meant juggling UI/UX and backend logic. I learned to manage my time effectively by focusing on one side at a time while gradually integrating both.

Working simultaneously on UI/UX design and backend logic was demanding. Switching between frontend and backend tasks often required careful planning and time management.

Time Management: Managing my time effectively and meeting deadlines was another challenge. However, breaking tasks down into smaller, manageable steps and tracking progress using tools like Trello helped me stay organized.

3.CONCLUSION

The Java Full Stack internship provided me with a comprehensive learning experience that strengthened both my technical skills and professional abilities. Throughout the project, I gained hands-on exposure to essential frontend technologies like HTML, CSS, JavaScript, and Tailwind CSS, along with practical knowledge of version control, hosting, and UI design tools. I also learned the importance of writing clean code, following best practices, and working with Agile methodologies. Professional Growth: I feel more confident in my technical abilities, having learned how to combine frontend and back-end technologies and having gained hands-on experience with the full stack web development lifecycle.

The website I contributed to, AIEDU ERP, provided a practical environment where I was able to apply and refine the skills I had learned throughout my internship. Working in a collaborative team setting, I actively participated in the development and testing phases of the platform, ensuring that it not only met the functional requirements but also provided an intuitive and seamless user experience. My contributions ranged from structuring web pages with HTML, styling them with CSS, and adding interactivity with JavaScript to testing the platform's performance and functionality using black-box testing techniques.

Overall, this internship enhanced my confidence in working on full-stack development tasks, improved my problem-solving abilities, and prepared me to handle real-time projects in a professional environment. The knowledge and experience gained during this period will play a significant role in my future career as a software developer

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